NAEEM NAIRO

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SUMMARY

Biomedical and mechatronics engineer with nearly 4 years of industry experience taking assistive mobility products from concept to mass production. Certified SOLIDWORKS Expert with strong expertise in CAD, FEA, DFM, GD&T, prototyping, and verification testing, plus sensor integration and embedded systems (ESP32, STM32). Focused on translating clinical and user needs into functional devices users can rely on every day.

ENGINEERING EXPERIENCE

AHL Logics **November 2021 to August 2025**

Product Design Engineer, Full Time *June 2022 to August 2025*

- **Full product development:** Led development of assistive mobility products (strollers, walkers, infant car seats) from concept through prototyping, testing, mass production and launch.
- **CAD and DFM:** Designed and optimized parts and assemblies in SOLIDWORKS, applying DFM principles to reduce cost, part count and manufacturing complexity.
- **Simulation and testing:** Ran structural FEA to improve strength and durability while reducing weight, then confirmed the predictions through physical prototype testing.
- **Engineering documentation:** Wrote design specifications, test plans and reports, technical drawings, BOMs, assembly instructions and manufacturing documentation.
- **Design transfer to production:** Worked with manufacturing, quality and suppliers to align drawings and BOMs and resolve DFM and tooling issues before production.

Product Design Engineer, Part Time *November 2021 to June 2022*

- **Concept development:** Contributed to mechanism design, material selection and prototype builds during concept refinement.
- **Quality and validation:** Took part in design reviews, prototype evaluation, FMEA, root cause analysis, verification and validation, and engineering changes with customers, suppliers and manufacturers.

University of New Haven **New Haven, CT**

Graduate Teaching Assistant, Instrumentation and Sensors Laboratory *Fall 2025 to Present*

- **Lab instruction:** Guide students through LabVIEW and NI ELVIS experiments in EMG signal acquisition and temperature sensing.
- **Rehabilitation device project:** Adapting a ball and beam control platform into a rehabilitation tool, with training strategies for stroke patients with mobility impairments.

EDUCATION

M.S. Biomedical Engineering | University of New Haven, Tagliatela College of Engineering
Relevant coursework: Biomedical Polymers; Biomedical Sensors and Instrumentation; Biostatistics; Biomechanics; Signal analysis.
B.S. Mechanical Engineering (Mechatronics) | PPU | January 2022
Relevant coursework: Mechanical design; Material science; Control system; Manufacturing processes; Mechatronic systems; Thermodynamic

TECHNICAL SKILLS

Quality & Regulatory: design verification and validation (V&V), FMEA, root cause analysis, ISO 13485 and FDA 21 CFR Part 820.
Medical & Assistive Technology: prosthetics, rehabilitation devices, mobility products, biomedical sensors, human centered design.
Analysis & Simulation: SOLIDWORKS Simulation, ANSYS, structural/static FEA, MATLAB, prototype testing and validation
Instrumentation & Embedded Systems: ESP32, STM32, I2C, LabVIEW, NI ELVIS, Arduino.
Design & Prototyping: SOLIDWORKS (CSWE), CAD modeling, DFM, GD&T.

CERTIFICATIONS

- Certified SOLIDWORKS Expert (CSWE), December 2025
- Certified SOLIDWORKS Professional Simulation, February 2025
- SOLIDWORKS Additive Manufacturing Associate, January 2025
- Additive Manufacturing: Materials for 3D Printing, December 2025

LEADERSHIP / AWARDS

- Fulbright Scholar, U.S. government international scholarship recipient, 2025 to 2027.
- Secretary, Biomedical Engineering Society, University of New Haven, Spring 2026 to Present; support events, professional coordination and member communications.
- President, Mechanical Engineering Society, Palestine Polytechnic University, Spring 2021.
- Professional affiliations: IEEE, ASME and Biomedical Engineering Society (BMES)
- SOLIDWORKS Champion, recognized for advanced proficiency in 3D CAD modeling, and product development.